Predictive Analysis of Crime Rates in Korea

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1. Introduction

Crime rates is routinely collected, maintained and analyzed by many nations worldwide. These statistics are essential for understanding public safety trends, allocating resources, and informing future policy decisions. Usually crime is examined in the context of social, economic, and demographic factors. However, some researches suggests perhaps unexpected influence could exist between climate and environmental conditions.

Recent studies have suggested a possible correlation between climatic variables and crime rates, particularly violent crimes. Although this connection may seem surprising at first, there is a growing body of evidence and several hypotheses that explore the biological and behavioral mechanisms that might underlie such a relationship. One of the theory focuses on the role of serotonin, a neurotransmitter known to regulate mood and aggressive behavior. Scientific research has shown that serotonin function may be weakened in high-temperature environments due to serotonin degradation. This reduction may result in increased secretion of hormones such as adrenaline and endorphins, both of which are associated with heightened aggression and impulsivity [2].

A research by Harvard School of Public Health suggests that higher temperatures can impair cognitive function and reduce problem-solving abilities [1]. In high-heat studying environment, students may find it more difficult to process information, control impulses, or make rational decisions, all of which are cognitive factors known to influence criminal activity. Taken together, these scientific findings point to a compelling possibility that crime rate can possible modeled and predicted through climate environmental data.

This project aims to explore within the context of South Korea, a country that experiences well-defined seasonal shifts—hot and humid summers contrasted by cold, dry winters. These climate extremes present a unique opportunity to examine the environmental determinants of crime in a diverse set of conditions. The South Korean government provides relevant datasets for the project, including accusation rate data from 2017 to 2019, made publicly available by the Prosecutors' Office through the national data repository (data.go.kr). Additionally, the Korean Statistical Information Service (KOSIS) offers extensive records of regional climate data over the same period.

This paper will utilize both crime and climate data to investigate the possibility of a predictive model for crime rate based on environmental variables. By integrating scientific theory, statistical evidence, and real world data, this study seeks to contribute to the broader discourse on how external, non-social environmental factors, such as weather, can influence human behavior and societal patterns. We will attempt 3 different modeling approaches; simple linear regression, auto-regressive moving average (ARMA), and Bayesian structural timeseries (BSTS). Through these several modeling approaches, we aim to better understand the dynamics at play and potentially offer insights that can inform future policy, urban planning, and public safety strategies.

2. Methods

2-1. Data Processing

Data for crime rate reported by each local police department can be found from various government database from Korea. Korean Statistical Information Service (KOSIS) and Public data repository (data.go.kr). Only crime types with $\geq 20,000$ are selected for the analysis.

Additionally some weather and financial data will be tested to see if they have any correlation with crime rate of each types. These data will be collected from kma.go.kr and index.go.kr respectively.

Finally, last 3 months were separated from the main aggregated data to be used as the test dataset.

2-2. Linear Regression

Simple linear regression models relationship between a climate variables ($y \in \mathbb{R}^n$) and each of the crime types ($X \in \mathbb{R}^{n \times p}$). Model can be shown as;

$$y = X\beta + \epsilon$$

where ($\beta \in \mathbb{R}^p$) is vector coefficient and $\epsilon \sim \mathcal{N}(0, \sigma^2 I)$ is Gaussian noise term.

The model was trained using scikit-learn LinearRegression() class. Dependent variables are standardized and model performance is evaluated using mean squared error (MSE).

2-3. Auto-Regressive Moving Average (ARMA)

ARMA models describe a time series using both autoregressive (AR) and moving average (MA). An ARMA(p, q) model can be defined as:

$$y_t = \sum_{i=1}^{p} \phi_i y_{t-i} + \sum_{j=1}^{q} \theta_j \epsilon_{t-j} + \epsilon_t$$

where ϕ_i is AR component (coeffcient), θ_j is MA component, and ϵ_t is noise term.

In this project ARMA(2,2) model was utilized with ARMA class from statsmodel package. Likewise MSE was calculated for evaluation.

2-4. Bayesian Structural Timeseries (BSTS)

BSTS model incorporates Bayesian inference with regression terms. For each time point t, model can be described as;

$$y_t = \mu_t + X_t^\top \beta + \epsilon_t$$

where μ_t is mean modeled with random walk, $beta \sim \mathcal{N}(0, \sigma_\beta^2 I)$ is vector of regression coefficients, and $\epsilon_t \sim \mathcal{N}(0, \sigma^2)$ is observed noise.

Posterior prediction is calculated using Markov Chain Monte Carlo (MCMC) and No-U-Turn Sampler (NUTS). These methods are implemented by PyMC package version 5. Posterior distribution can be shown as;

$$p(\hat{y}_t \mid \text{data}) = \int p(y_t \mid \mu_t, \beta, \sigma) \cdot p(\mu_t, \beta, \sigma \mid \text{data})\, d\mu_t\, d\beta\, d\sigma$$

and posterior predictive mean for observed data is;

$$\hat{y}_t = \mathbb{E}_{p(\mu_t, \beta)}[\mu_t + X_t^\top \beta]$$

Finally we calculated MSE for comparison with other models.

3. Results

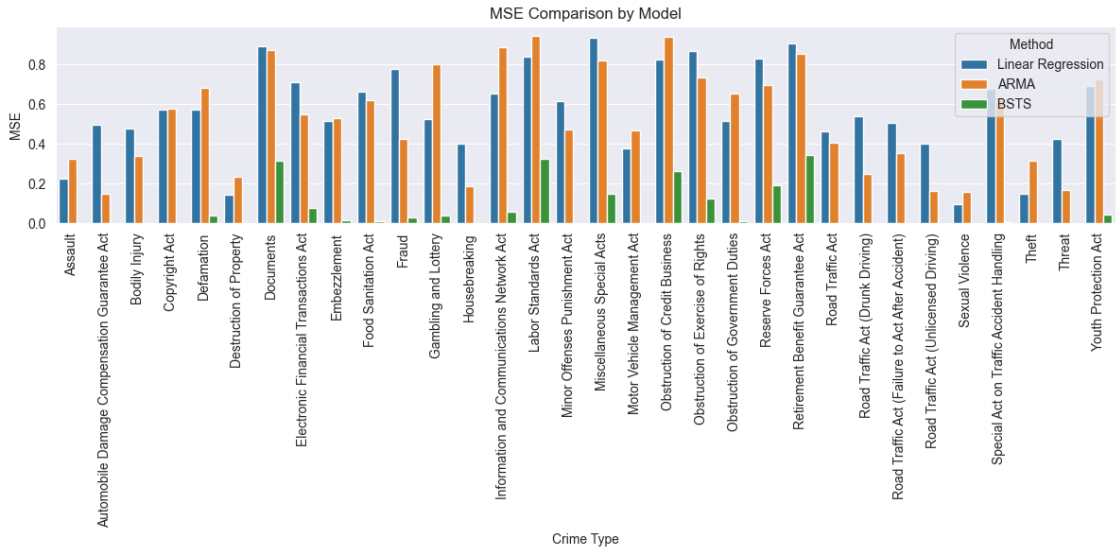

Figure 1: Summary of MSE for each crime types on training dataset

We evaluated the performance of three predictive models (Linear Regression, ARMA, and BSTS) using MSE across all crime categories. Initially, each model was trained on the full dataset and evaluated on the same training set (entire 3 year period). The BSTS model outperformed both the Linear Regression and ARMA models, which showed similar performance levels. Figure 1 shows the distribution of MSE values for each model across different crime types. While the BSTS model demonstrated superior performance, note that this was trained and evaluated on the same dataset, inevitably resulting in overfitting. This suggests that the model might have captured noise rather than underlying patterns in the data. Therefore, while BSTS provided better results on the training set, its generalization to future datasets remains uncertain without further cross-validation.

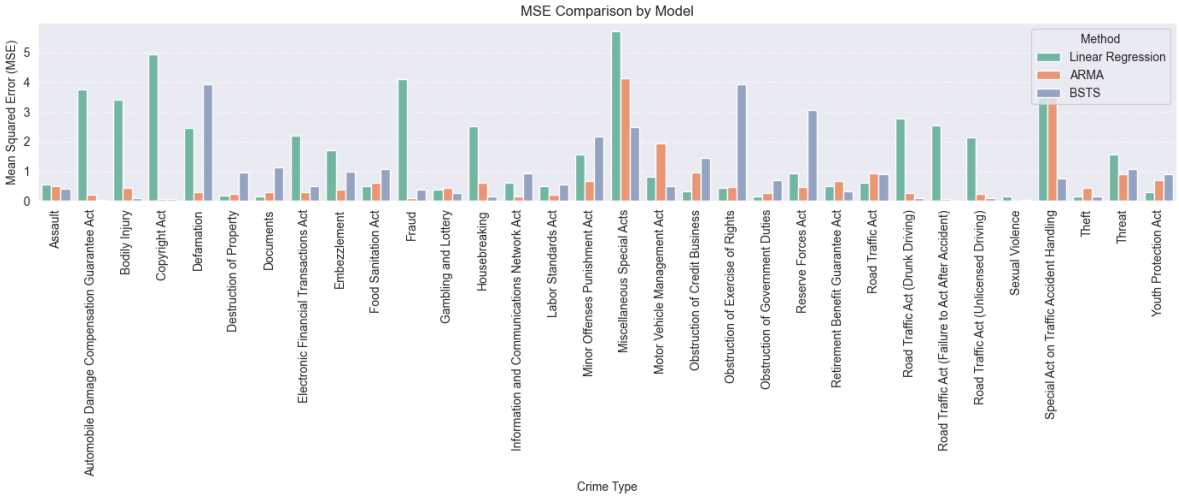

Figure 2: Summary of MSE for each crime types on training dataset

To assess the models' generalizability, we tested them on a three-month test dataset. We trained the

model for 33 month period and excluded last 3 month for evaluation. This time, the BSTS model did not perform as well, suggesting it had overfitted to the training dataset in previous attempt. In contrast, ARMA performed notably better than the other models, making it the most promising candidate for future predictions. However because all models were trained exclusively on climate data, the overall predictive power was inherently limited. Socioeconomic, demographic, and other external factors are not included in this study. Therefore, some level of overfitting is expected across all three modeling approaches.

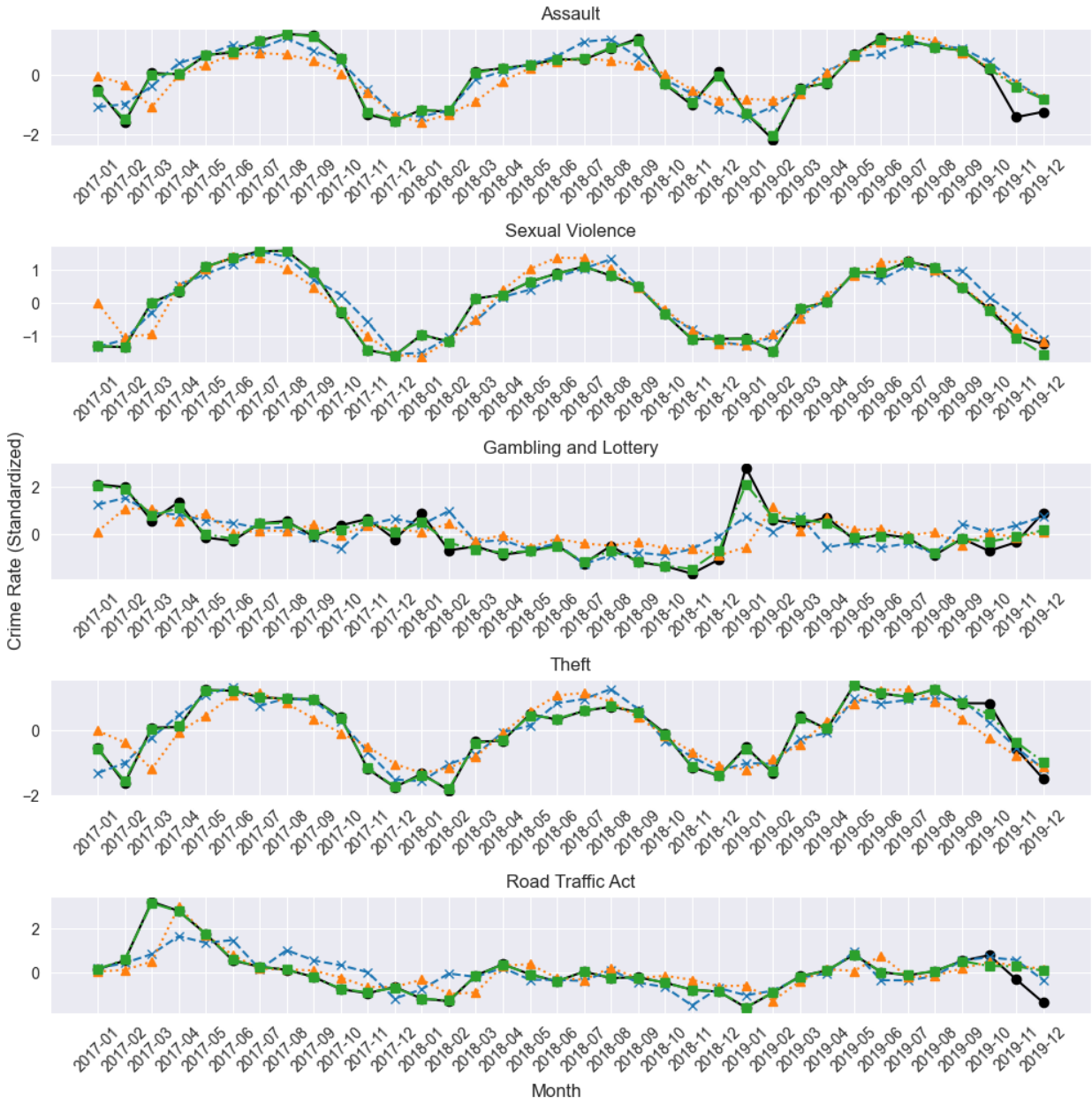

Figure 3: Prediction of most predictable 5 crime types.

To further understand model behavior, we ranked crime categories by their average prediction error. The five crime categories with the lowest MSE (i.e., the most accurately predicted) were sexual violence, destruction of property, theft, assault, and unlicensed driving, as shown in Figure 3. Consistent with previous studies, crimes that involve direct physical or interpersonal aggression tended to be more predictable, suggesting a

potential link between violent behavior and climatic factors, such as heat or humidity.

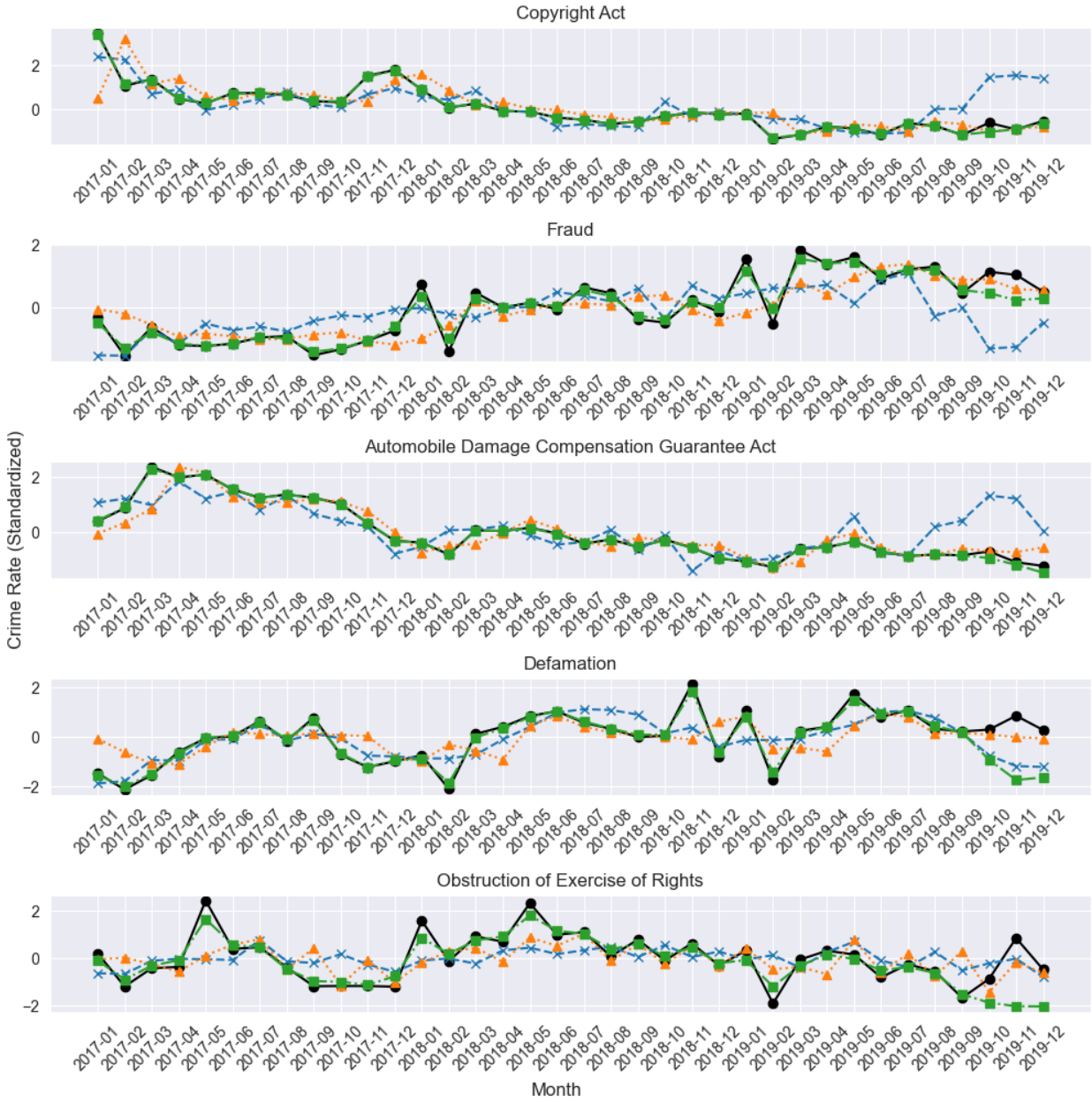

Figure 4: Prediction of least predictable 5 crime types.

In contrast, figure 4 shows the five crime categories with the highest MSE (the most difficult to predict), which include fraud, gambling, miscellaneous crimes, information and communication violations, and obstruction of the exercise of rights. Additionally we can see for Jan 2017 to Oct 2019, which were period used for train and fit models, models are overfitted for this period in general compared to last 3 months which is test dataset.

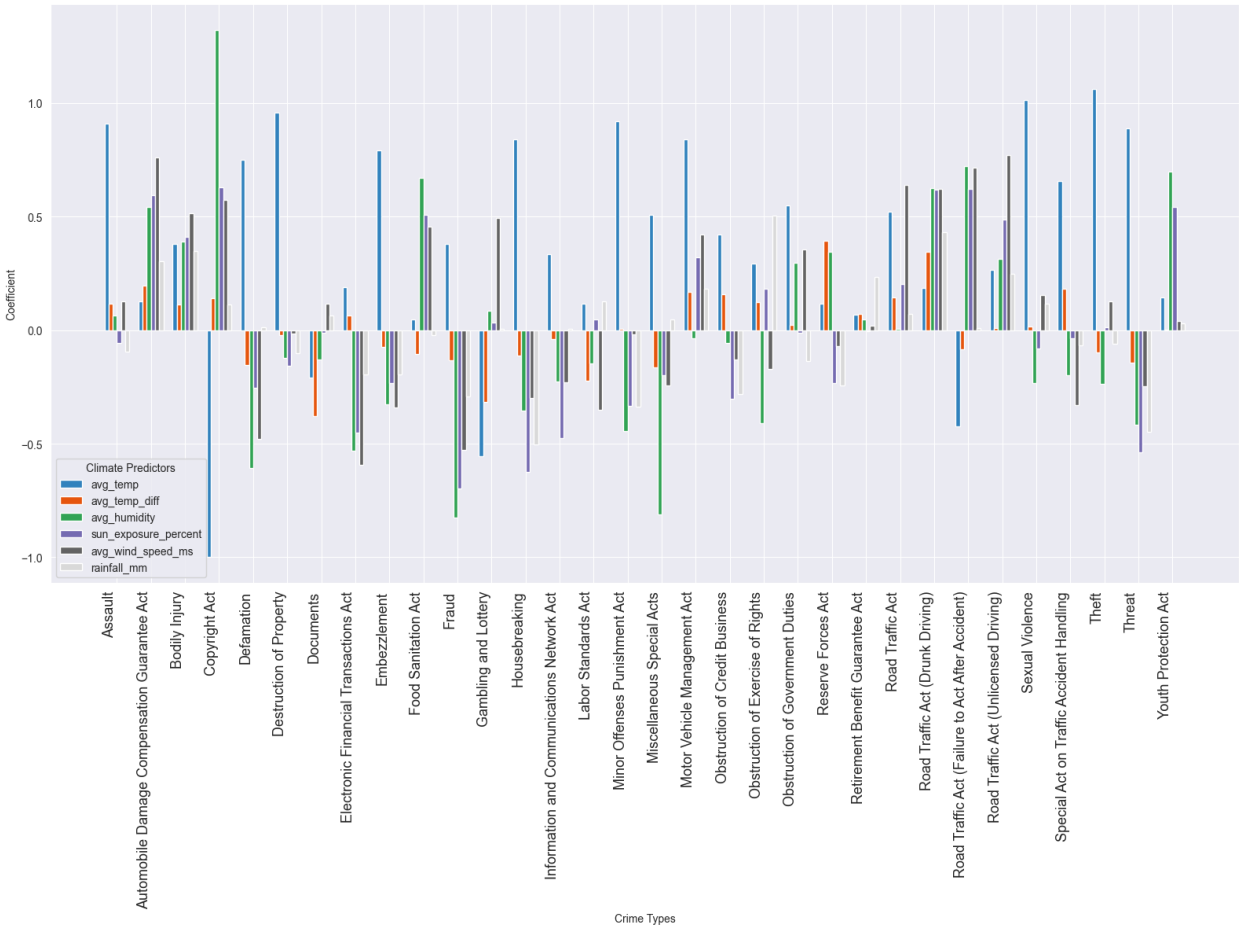

Figure 5: Summary of co-efficient from linear regression.

Lastly, we examined the coefficients of the Linear Regression model to assess the influence of each climate variable on crime rates. As foreshadowed from previous researches, violent crime types showed a strong positive correlation with high temperatures. However, the influence of the other climate variables was less consistent, which was somewhat underwhelming. The study had expected to observe a more pronounced relationship between humidity and violent crime, but this was not evident in the results. Yet small interesting finding was observed; such as increased drunk driving cases when there is high humidity and rainfall, suggesting people tends to risk drunk driving under heavy weather.

4. Discussion

In this study, we compared three predictive models for crime forecasting using climate data. Overall, the BSTS model have shown strong tendency to overfit to noise. In general, the ARMA model which relies on historical data patterns and does not involve complex temporal relationships like BSTS, performed relatively better on the test dataset, making it a more reliable model for future predictions.

A limitation of our study remains evident. External datasets such as socioeconomic or demographic are not incorporated or compared to climate variables in this project. For instance, research has shown that factors such as income inequality, unemployment, and social policies can influence crime rates. The absence of these variables likely limited the predictive power of all three models and resulted in overfitting to the noise. In future research, it would be important to incorporate such data to improve usefulness of the models. The inclusion of these factors might will help mitigate some of the overfitting observed.

Additionally, the results of this study suggest that violent crimes, such as assault and sexual violence, were more predictable than non-violent crimes, possibly due to their stronger correlation with environmental factors like heat. This finding is consistent with previous research suggesting that warmer temperatures may increase the likelihood of violent behavior. However, other climate variables, such as humidity, did not show consistent effects.

The inclusion of further external factors could provide deeper insights into the dynamics between climate and crime. Incorporating data on public health, such as hospital admissions related to heat stress or alcohol-related incidents, might also offer a clearer understanding of the mechanisms through which climate influences crime. Furthermore, a crime type specific approaches would provide more constructive model as seen in the case of drunk driving. Overall, this project was limited in terms of its capability and generalization, yet able to provide promising possibility of crime forecasting model based on climate variable.

References

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[2] Paul J Schwartz, Thomas A Wehr, Norman E Rosenthal, John J Bartko, Dan A Oren, Christopher Luetke, and Dennis L Murphy. Serotonin and thermoregulation. *Neuropsychopharmacology*, 13(2):105–115, 1995.